



University of Wisconsin
SCHOOL OF MEDICINE
AND PUBLIC HEALTH

BMI/STAT-768

Statistical Methods for Medical Image Analysis and High Dimensional Data

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<https://github.com/laplcebeltrami/BMI768>

Unit Objectives

- 1) Understand curve & surface parametrization
- 2) Understand Riemannian metric tensors
- 3) Understand geodesics

Geometric Data Analysis

Geometric Data Analysis (GDA) is a field that focuses on understanding and interpreting the geometric structures inherent in complex datasets. It leverages mathematical and statistical techniques to explore properties such as shape, curvature, and topology, enabling the identification of patterns and relationships within the data. GDA encompasses methods like manifold learning, principal component analysis, and dimensionality reduction, which help in visualizing high-dimensional data in more interpretable forms.

Information
geometry

Data embedding

Covariance
matrix

Data manifolds

GDA

quantifies
the shape
of data

Symmetric positive
definite matrices

Statistics on
manifolds

PCA

Spectral clustering

Functional-PCA

Spectral geometry

Geometric Data Analysis

Geometric Data Analysis (GDA) is a field that focuses on understanding and interpreting the geometric structures inherent in complex datasets. It leverages mathematical and statistical techniques to explore properties such as shape, curvature, and topology, enabling the identification of patterns and relationships within the data. GDA encompasses methods like manifold learning, principal component analysis, and dimensionality reduction, which help in visualizing high-dimensional data in more interpretable forms.

References

Widely used
basic textbook
on differential
geometry

Differential Geometry of Curves and Surfaces

Manfredo P. do Carmo

*Instituto de Matematica Pura e Aplicada (IMPA)
Rio de Janeiro, Brazil*



Required reading

Chung et al. 2022, Statistical Analysis on Brain Surfaces, [arXiv:2203.06665](https://arxiv.org/abs/2203.06665)

Differential Geometry of a Curve

A **smooth curve** in 2D can be described by a parametric function

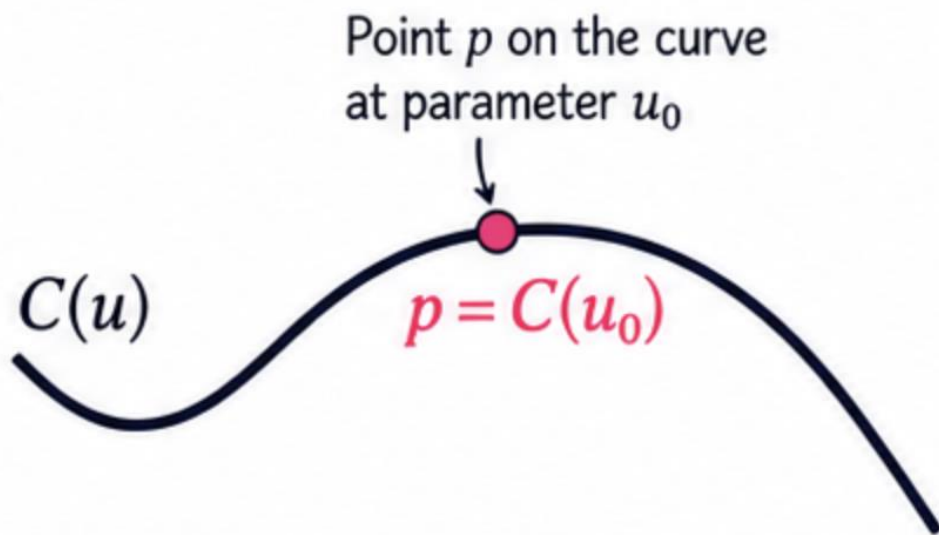
$$C(u) = (x(u), y(u))$$

u : parameter (e.g., arc length or any local coordinate)

Example: a circle

$$x(u) = \cos(u)$$

$$y(u) = \sin(u) \quad 0 \leq u < 2\pi$$



Objects must be parameterized
→ Establish local coordinates.

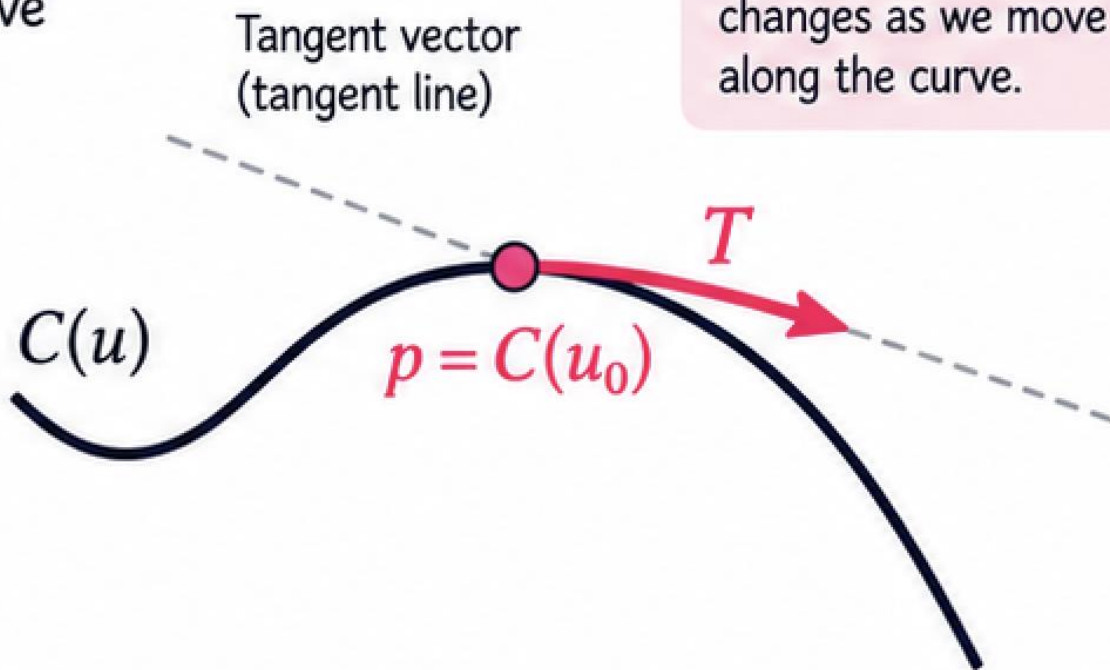
Tangent Vector

The tangent vector points to the direction of the curve at u_0 .

$$C_u = \frac{\partial C(u)}{\partial u}$$

Unit tangent vector:

$$T = \frac{C_u}{\|C_u\|}$$



T points in the direction of the curve. It shows how the position changes as we move along the curve.

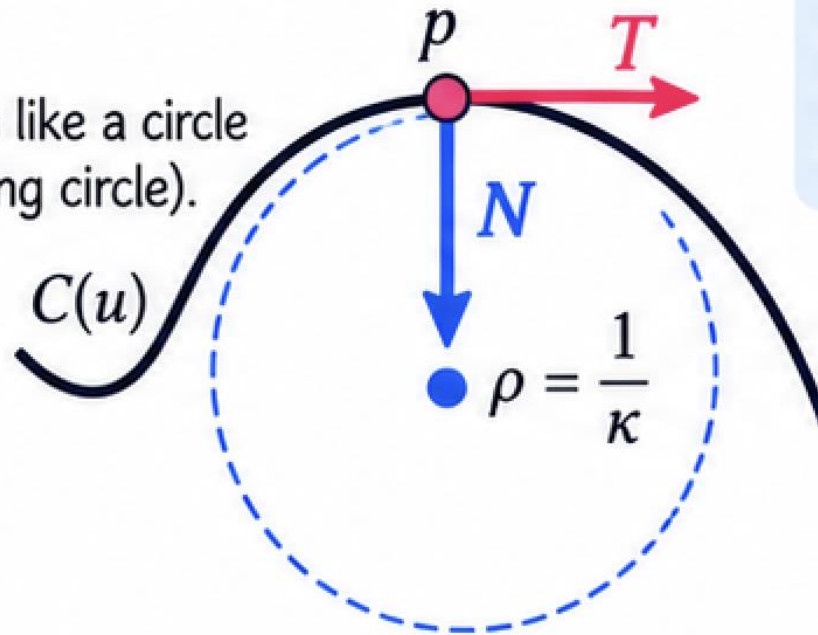
Exercise. Work out tangent vector computation for a circle parametrized by $x(u) = \cos(u)$, $y(u) = \sin(u)$.

Curvature of a 1D Curve

A 1D curve has only one curvature κ at u_0 .

The curve locally looks like a circle with radius ρ (osculating circle).

$$\kappa = \frac{1}{\rho}$$



The osculating circle is the circle that best approximates the curve near p . Its radius ρ is the inverse of the curvature κ .

Curvature measures how fast the curve bends.

Exercise. Determine the curvature as a function of $C(u)$ numerically in Matlab for few synthetic example.

White matter fibers

James Gee
Univ. Penn

Anterior (Front)

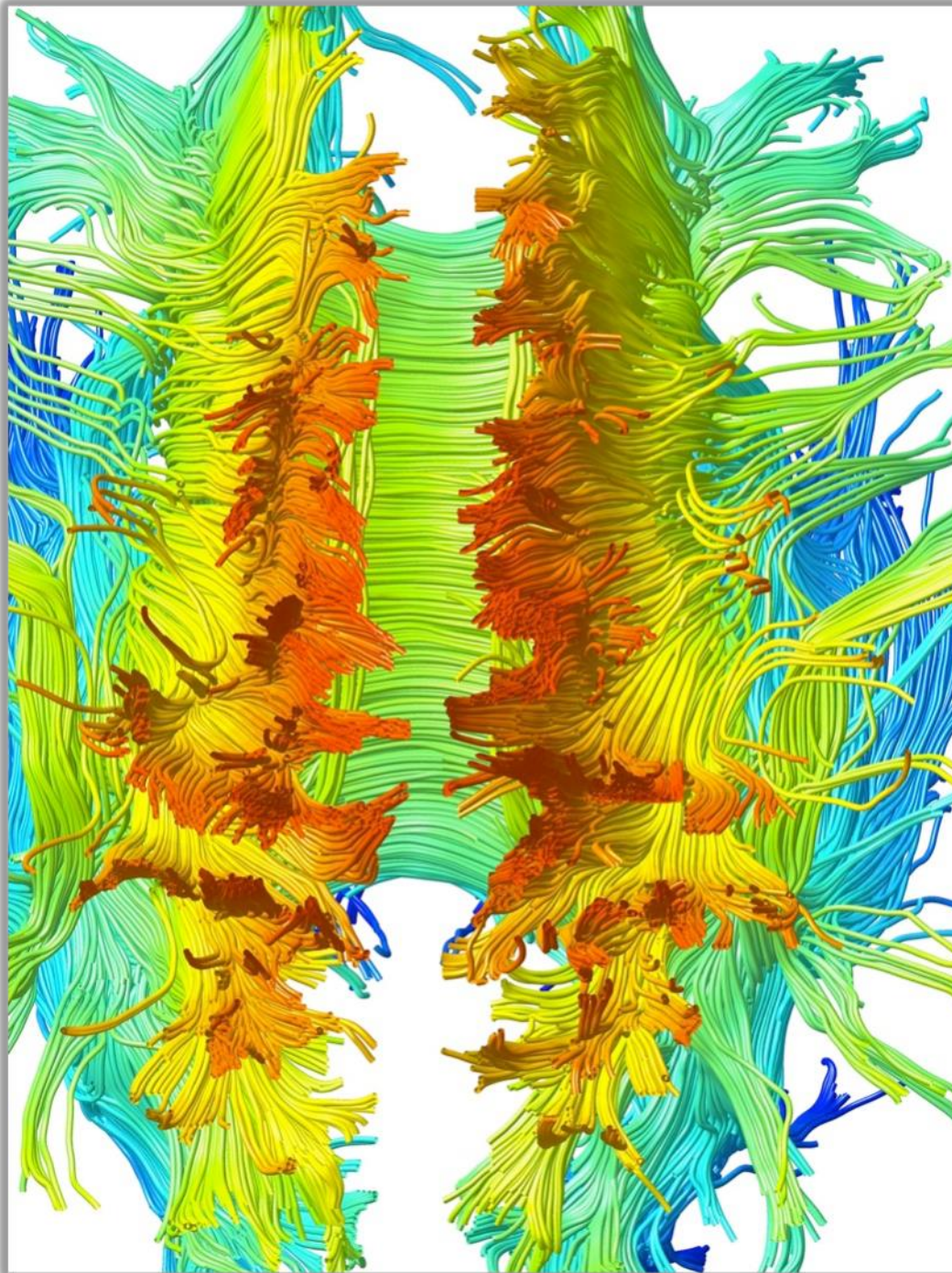
side view

top view

Posterior (Back)

Fibers passing through
the splenium of the
corpus callosum



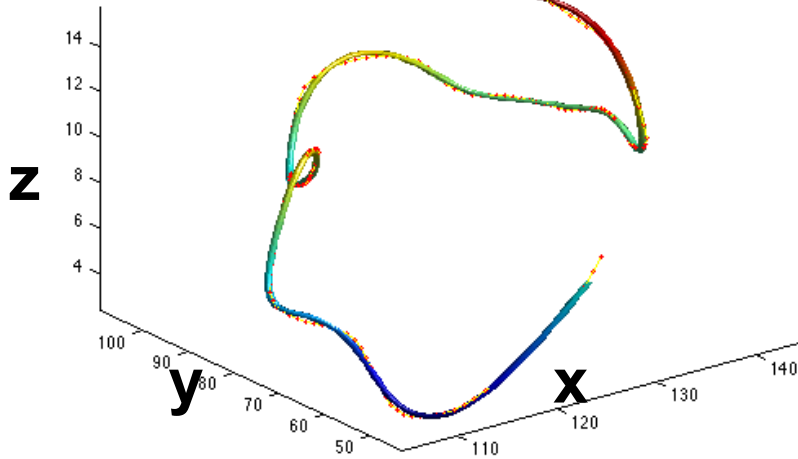


*We used cosine basis to
parameterize 3D curves*

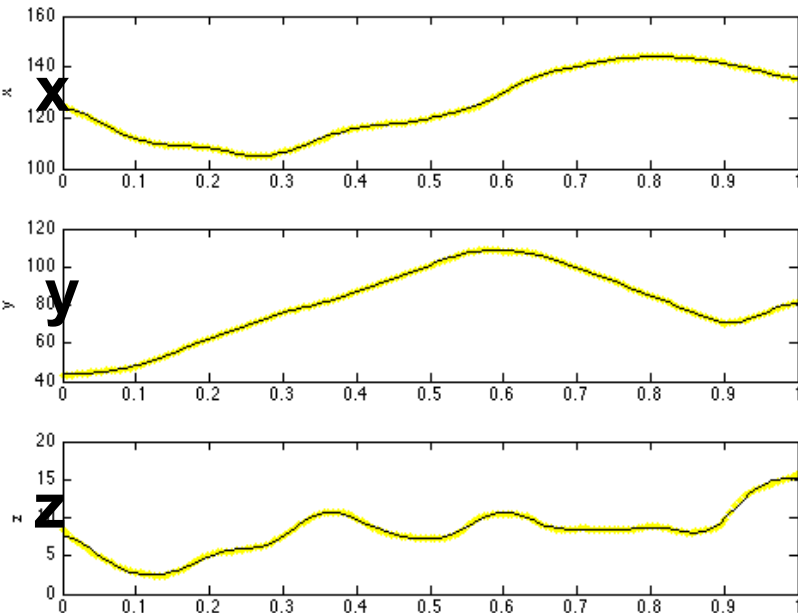
*Cover art from book
Chung, M.K. 2012
Computational Neuroanatomy:
The Methods, World Scientific
Press*

Matlab visualization

White matter fiber tract



parameterization



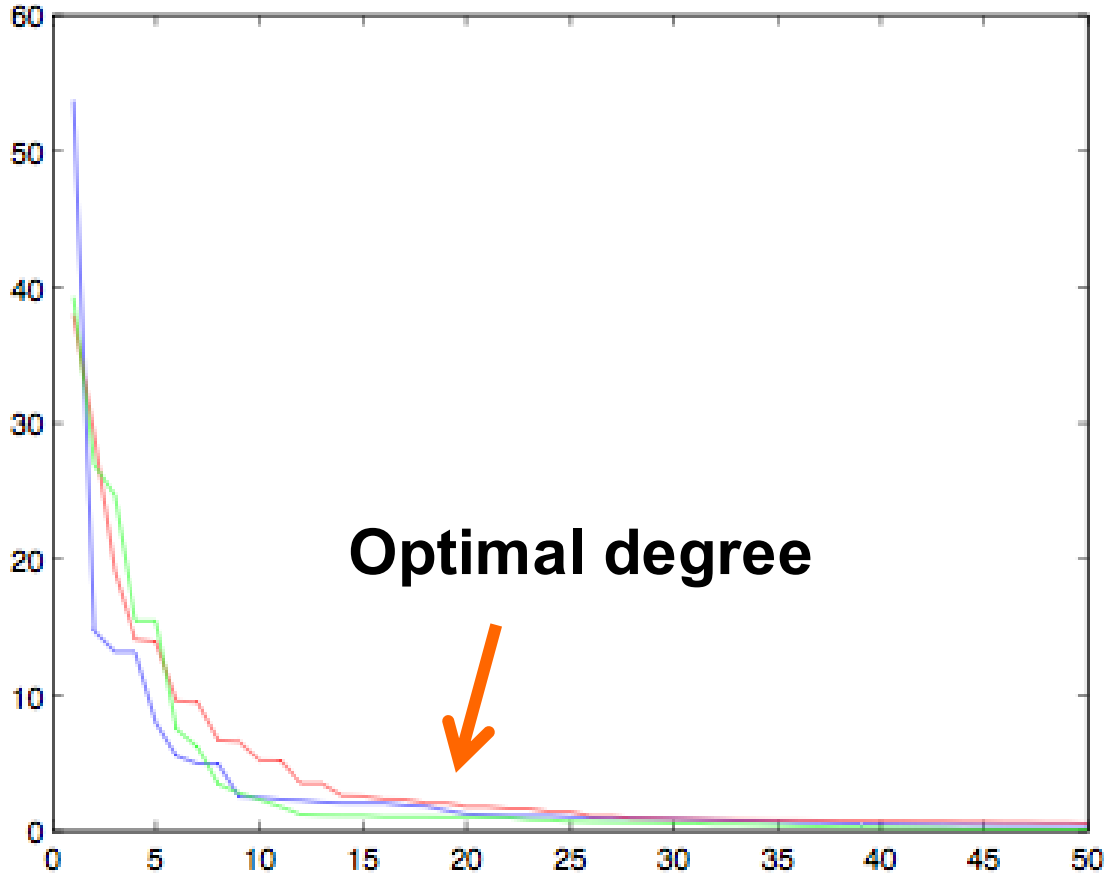
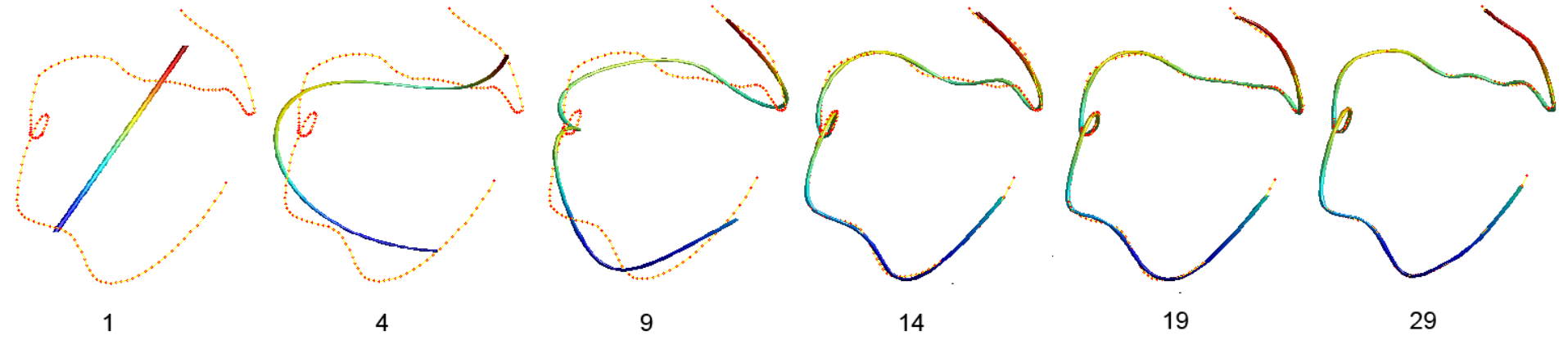
88.1799	56.6336	5.7367
-12.4775	-11.2552	-2.0791
2.4336	-15.4428	-0.4021
4.3956	2.2733	-0.9354
-0.0106	-0.0674	0.6999
2.1773	-2.4194	-0.1176
0.5808	0.8390	1.2942
0.0615	-0.1893	0.1188
-0.2629	0.7524	0.1089
0.7909	-0.7276	-0.1901
0.5458	0.6236	0.6939
0.4295	-0.4337	0.2185
0.2150	0.4157	0.0254
0.1584	-0.1973	0.0762
-0.1557	0.2466	-0.1086
0.0632	-0.0978	-0.0208
0.0389	-0.0143	-0.0284
-0.0014	-0.1193	0.1970
0.0004	0.0129	-0.0198
0.1342	0.0002	0.0260

Any tract can be compactly parameterized with about 60 coefficients.

basis expansion

$$\rightarrow (x, y, z)' = \sum_{l=0}^{19} \beta_l \cos(l\pi t)$$

Cosine series representation



The optimal degree chosen using the forward model selection.

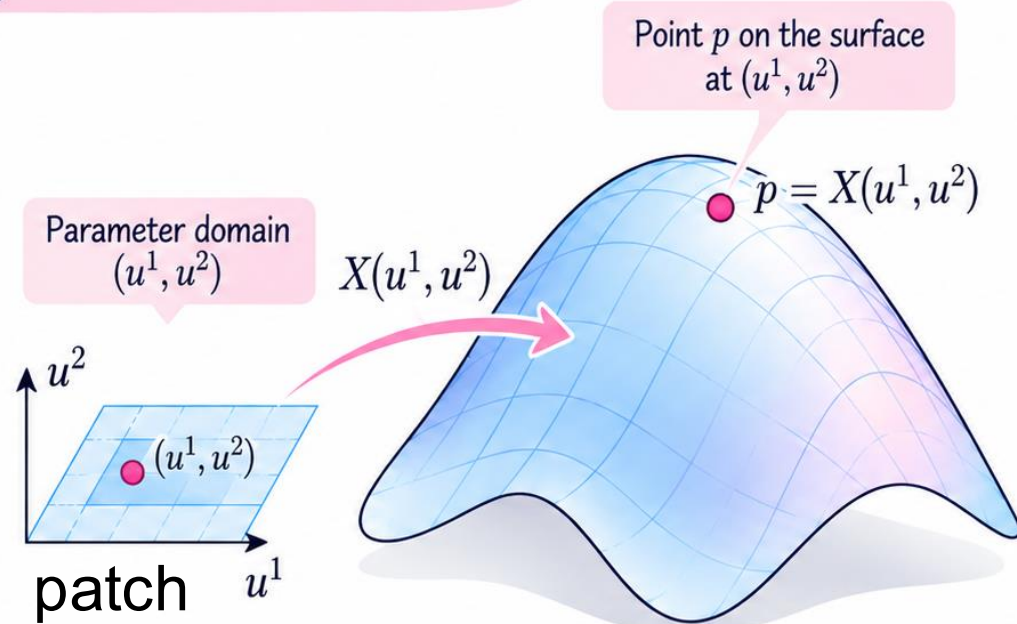
Exercise. Estimate curvature and display on top of the 3D curve.

Differential Geometry of a Surface

A **smooth surface** in 3D can be described by a parameterization

$$X(u^1, u^2) = (x(u^1, u^2), y(u^1, u^2), z(u^1, u^2))$$

(u^1, u^2) : local coordinates
on a small patch (parameter domain)



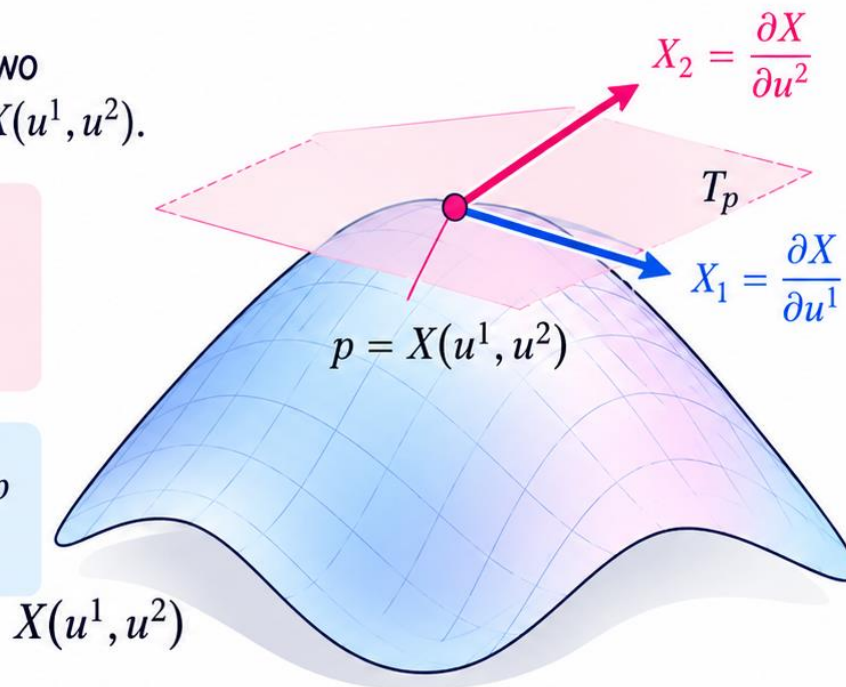
A parameterization gives a **local coordinate system** on the surface, allowing us to study its geometric properties using calculus.

Tangent Vectors on a Surface

The partial derivatives of X give two **tangent vectors** at the point $p = X(u^1, u^2)$.

$$X_1 = \frac{\partial X}{\partial u^1}, \quad X_2 = \frac{\partial X}{\partial u^2}$$

X_1 and X_2 lie in the tangent plane at p and span the local tangent space.



The tangent plane at p is the plane spanned by X_1 and X_2 .

These vectors describe the directions in which we can move on the surface from the point p .

Example: quadratic surface

$$x = u^1$$

$$y = u^2$$

$$z = \beta_0 + \beta_1 u^1 + \beta_2 u^2 + \beta_3 u^1 u^2 + \beta_4 (u^1)^2 + \beta_5 (u^2)^2$$

$$X(u^1, u^2) = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} u^1 \\ u^2 \\ \beta_0 + \beta_1 u^1 + \dots + \beta_5 (u^2)^2 \end{pmatrix}$$

$$\frac{\partial X}{\partial u^1} = \begin{pmatrix} 1 \\ 0 \\ \beta_1 + \beta_3 u^2 + 2\beta_4 u^1 \end{pmatrix}$$

At origin $(u^1, u^2) = (0, 0)$,

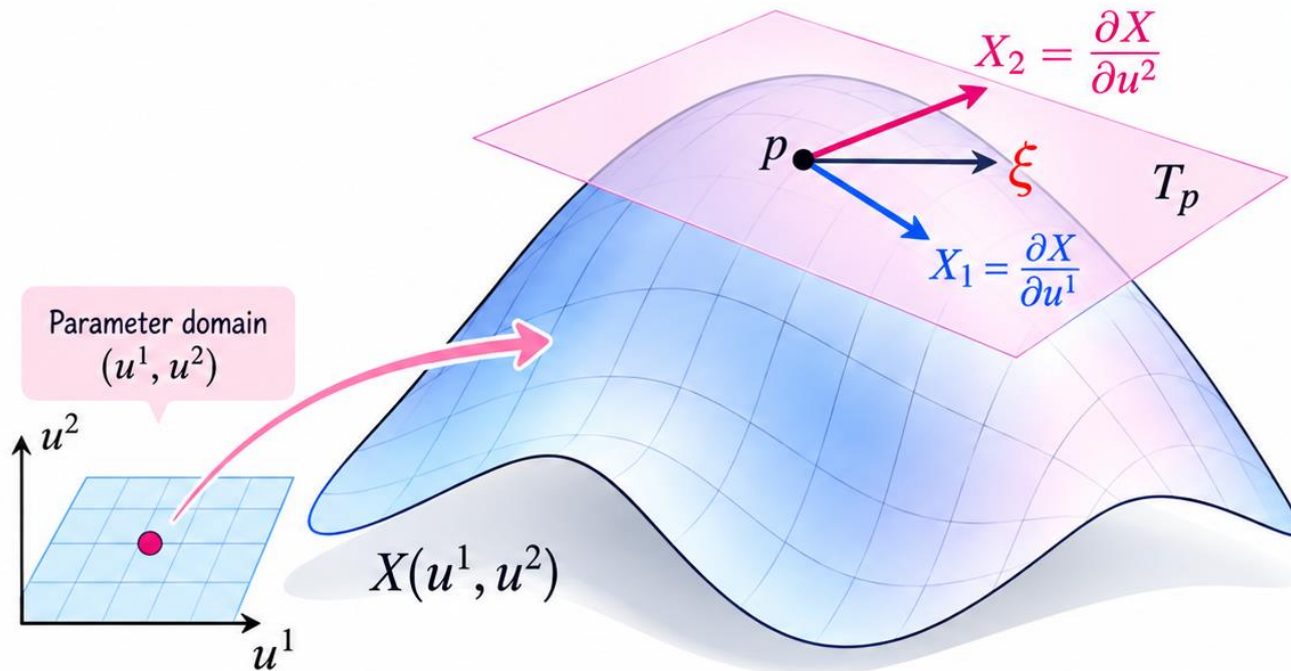
$$\frac{\partial X}{\partial u^2} = \begin{pmatrix} 0 \\ 1 \\ \beta_2 + \beta_3 u^1 + 2\beta_5 u^2 \end{pmatrix}$$

$$\frac{\partial X}{\partial u^1} = \begin{pmatrix} 1 \\ 0 \\ \beta_1 \end{pmatrix}$$

$$\frac{\partial X}{\partial u^2} = \begin{pmatrix} 0 \\ 1 \\ \beta_2 \end{pmatrix}$$

Tangent Plane to a Surface

The **tangent plane** T_p at a point p is the plane spanned by the two tangent vectors X_1 and X_2 .



Tangent plane

$$\xi \in T_p$$

$$\xi = u^1 X_1 + u^2 X_2$$

T_p is the set of all tangent vectors at p , i.e., all linear combinations of X_1 and X_2 .

First Fundamental Form

Differential form

$$d\xi = du^1 X_1 + du^2 X_2$$

Squared length (Riemannian metric)

$$d\xi^2 =^d \langle d\xi, d\xi \rangle = \sum_{i,j} \langle X_i, X_j \rangle du^i du^j$$

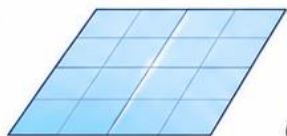
Metric tensor

$$g_{ij} = \langle X_i, X_j \rangle \quad (i, j = 1, 2) \quad \longrightarrow \quad g = \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix}$$

Examples of surfaces and metric tensor

Plane $\mathbb{R}^2$

$$X(u^1, u^2) = (u^1, u^2, 0)$$



$$g_{ij} = \delta_{ij}$$

(Euclidean metric)

Sphere S^2 (radius R)

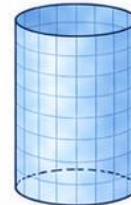
$$X(\theta, \phi) = (R \sin\theta \cos\phi, R \sin\theta \sin\phi, R \cos\theta)$$



$$g = \begin{pmatrix} R^2 & 0 \\ 0 & R^2 \sin^2 \theta \end{pmatrix}$$

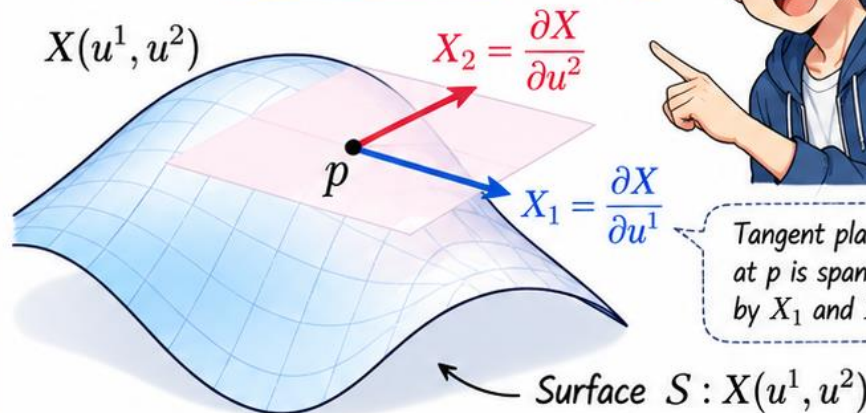
Cylinder (radius R)

$$X(u, v) = (R \cos u, R \sin u, v)$$



$$g = \begin{pmatrix} R^2 & 0 \\ 0 & 1 \end{pmatrix}$$

The first fundamental form measures lengths and angles on the surface.



Tangent plane at p is spanned by X_1 and X_2 .

The first fundamental form gives the inner product of tangent vectors, so we can compute lengths and angles *intrinsic to the surface*!



The first fundamental form encodes the *intrinsic geometry* of the surface.



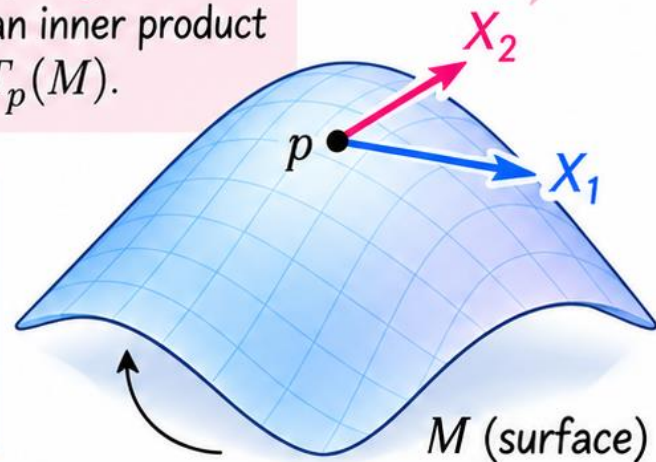
Riemannian metric tensor



To measure lengths and angles on a surface, we need a **metric**!

A **Riemannian manifold** (M, g) is a smooth manifold M with an inner product g on each tangent space $T_p(M)$.

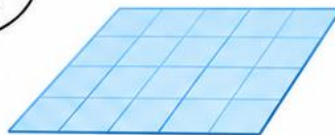
At each point p , the metric **varies smoothly** so that if X_1 and X_2 are differentiable tangent vector fields, then $g(X_1, X_2)$ is a smooth function on M .



The family of inner products g_p on all tangent spaces is called the **Riemannian metric tensor**.

If the metric doesn't vary (i.e., g is constant), we just get the usual **Euclidean space**.

No curvature, no surprises... just flat space!



$$g_{ij} = \delta_{ij}$$

(Euclidean metric)

When the metric changes from point to point, we capture the **intrinsic geometry** of the surface (curvature, etc.)!



Example: quadratic surface

$$x = u^1$$

$$y = u^2$$

$$z = \beta_0 + \beta_1 u^1 + \beta_2 u^2 + \beta_3 u^1 u^2 + \beta_4 (u^1)^2 + \beta_5 (u^2)^2$$

$$X_1 = \begin{pmatrix} 1 \\ 0 \\ \beta_1 \end{pmatrix} \quad X_2 = \begin{pmatrix} 0 \\ 1 \\ \beta_2 \end{pmatrix}$$

$$g_{11} = \langle X_1, X_1 \rangle = \|X_1\|^2 = 1 + \beta_1^2$$

$$g_{12} = \langle X_1, X_2 \rangle = \beta_1 \beta_2$$

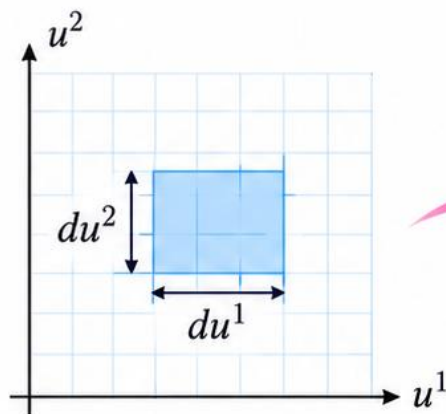
$$g_{22} = \langle X_2, X_2 \rangle = \|X_2\|^2 = 1 + \beta_2^2$$

$$\rightarrow g = \begin{pmatrix} 1 + \beta_1^2 & \beta_1 \beta_2 \\ \beta_1 \beta_2 & 1 + \beta_2^2 \end{pmatrix}$$

positive
definite
symmetric

Area element on a surface

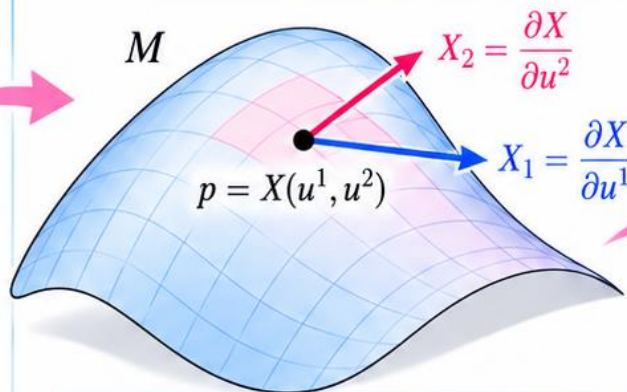
1 Parameter domain $N \subset \mathbb{R}^2$



A small rectangle in the parameter space has area $du^1 du^2$.

2 Parametrization $X: N \rightarrow M$

$$X(u^1, u^2) = (x(u^1, u^2), y(u^1, u^2), z(u^1, u^2))$$



The rectangle maps to a curved patch spanned by the tangent vectors X_1, X_2 .

A small patch $du^1 du^2$ in the parameter domain becomes a small curved patch on the surface. Its area is scaled by the **Jacobian determinant**!

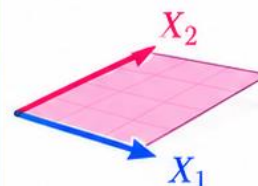


3 Area element (induced metric)

The area of the small patch is

$$d\mu(p) = \sqrt{\det g} du^1 du^2$$

where $g_{ij} = \langle X_i, X_j \rangle$ is the Riemannian metric (first fundamental form).



Area of the parallelogram spanned by X_1, X_2 is $\sqrt{\det g} du^1 du^2$.

Connection to the Jacobian determinant

The derivative dX (Jacobian matrix) is

$$dX = \begin{pmatrix} \frac{\partial x}{\partial u^1} & \frac{\partial x}{\partial u^2} \\ \frac{\partial y}{\partial u^1} & \frac{\partial y}{\partial u^2} \\ \frac{\partial z}{\partial u^1} & \frac{\partial z}{\partial u^2} \end{pmatrix} = (X_1 \quad X_2)$$

This is like the Jacobian matrix in change of variables, but now the columns are tangent vectors in $\mathbb{R}^3$.

Area scaling = length of cross product

$$\begin{aligned} d\mu &= \|X_1 \times X_2\| du^1 du^2 \\ &= \sqrt{\langle X_1, X_1 \rangle \langle X_2, X_2 \rangle - \langle X_1, X_2 \rangle^2} du^1 du^2 \\ &= \sqrt{\det g} du^1 du^2 \end{aligned}$$

Just like $|\det J|$ in change of variables, but adapted to the surface!



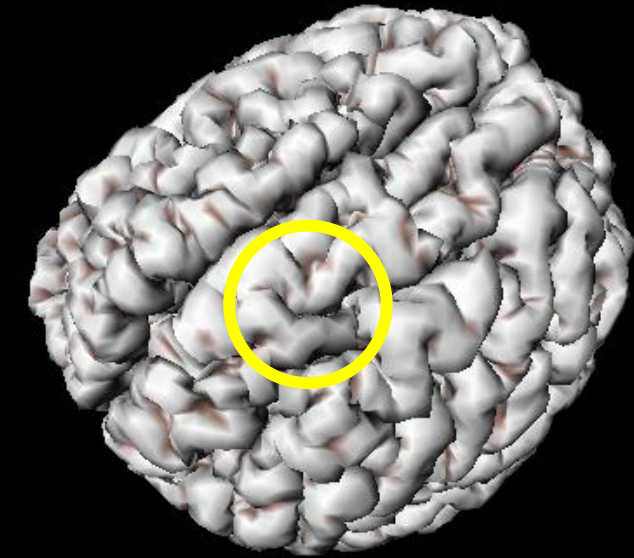
Total area of M

$$\begin{aligned} \text{Area}(M) &= \int_M d\mu(p) \\ &= \int_N \sqrt{\det g} du^1 du^2 \end{aligned}$$

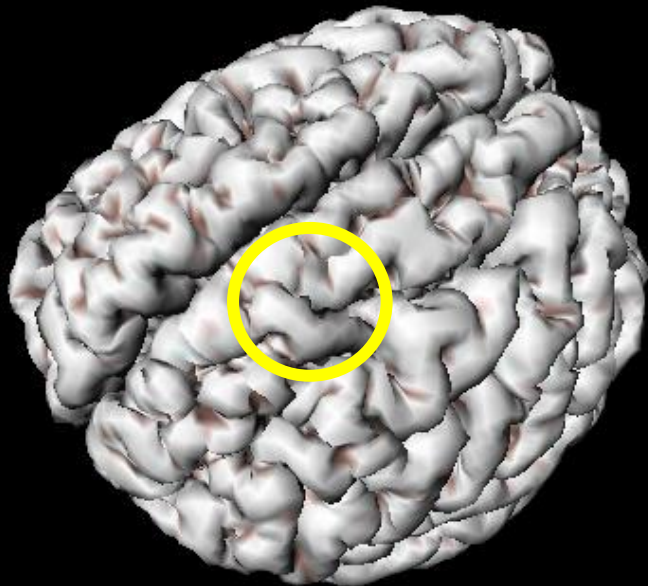
We integrate the local area element over the parameter domain, just like in the change of variables formula $\int_{\Omega} f(x) |\det J| dx$!

The **Jacobian determinant** measures how the parameter patch is stretched on the surface.

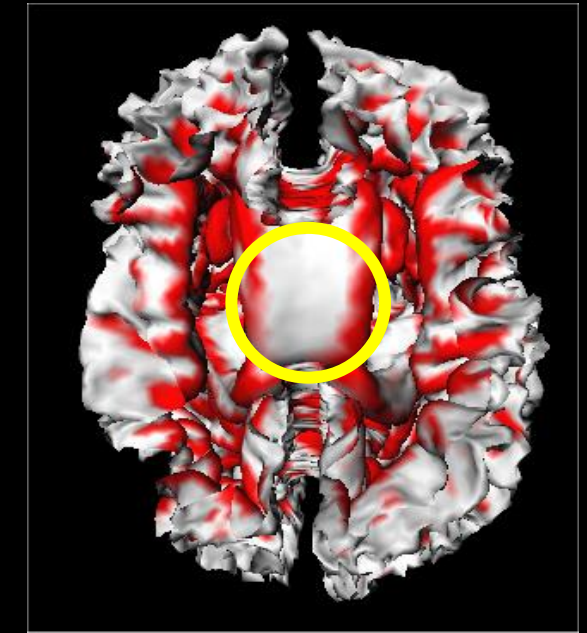
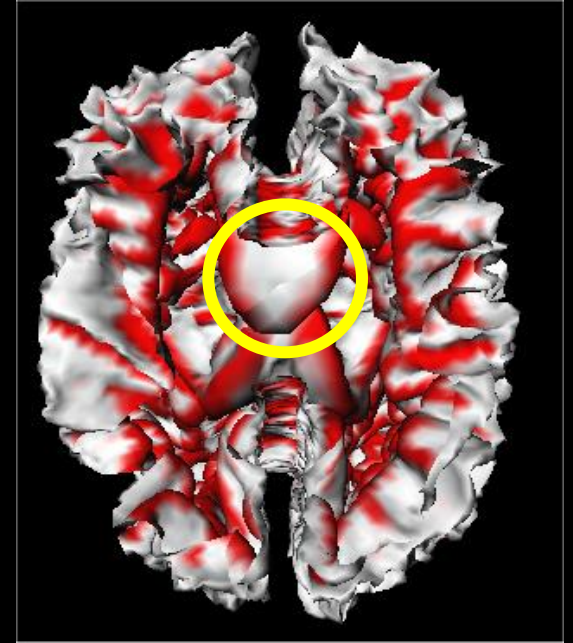
How to quantify brain surface growth and atrophy?



14 year old



19 year old



Tensor-based morphometry

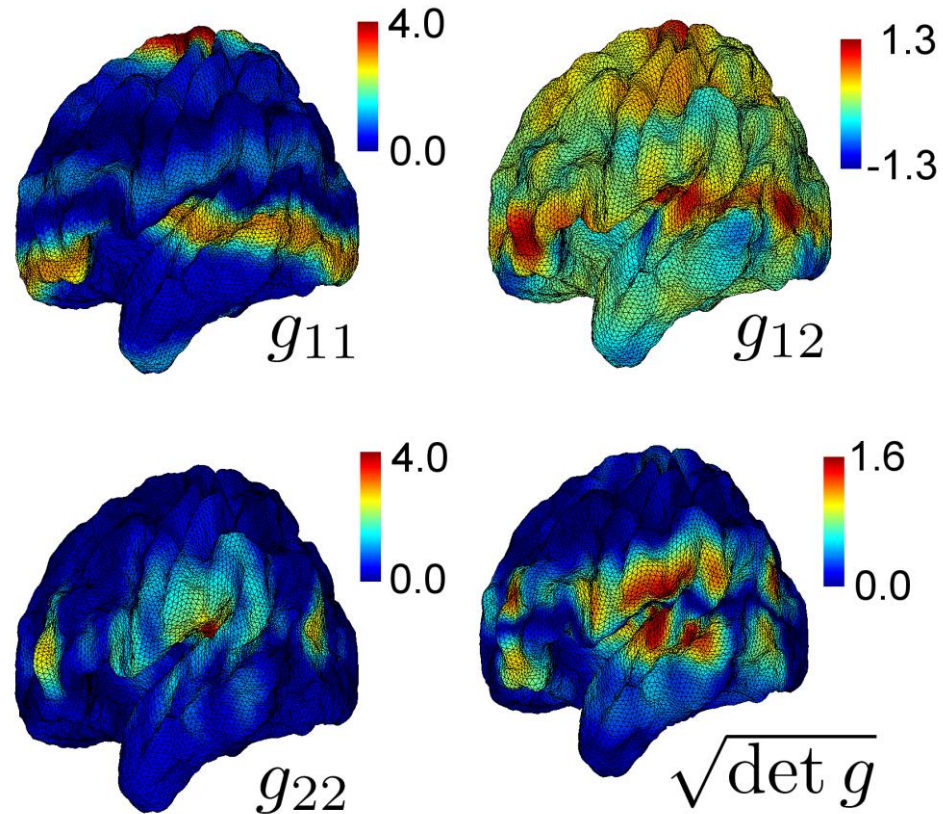
Exercise: Show surface area is invariant under parameterization

Surface parameterization:

$$u = (u^1, u^2) \mapsto X(u)$$

Riemannian metric tensor:

$$g_{ij} = \left\langle \frac{\partial X}{\partial u^i}, \frac{\partial X}{\partial u^j} \right\rangle$$

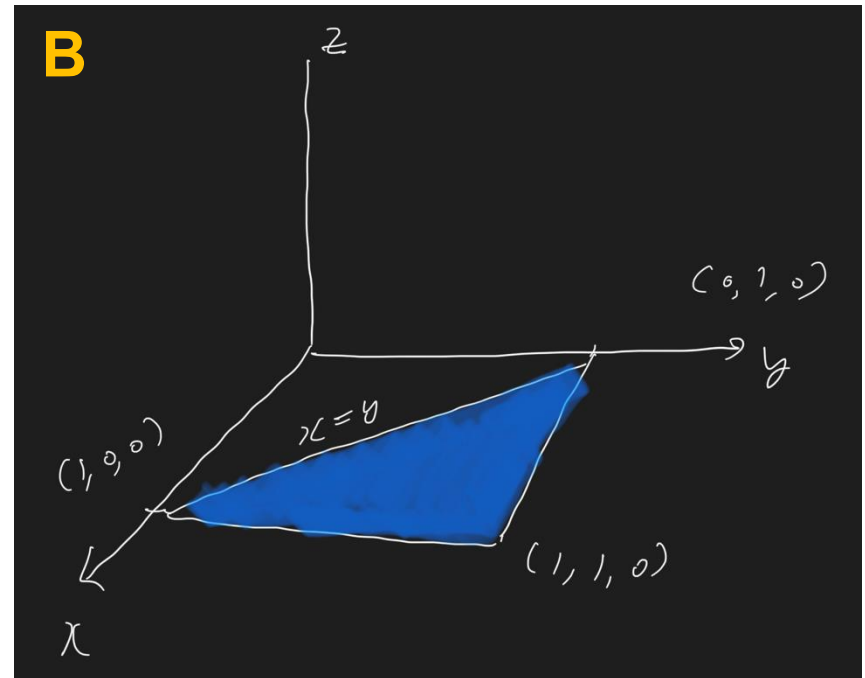
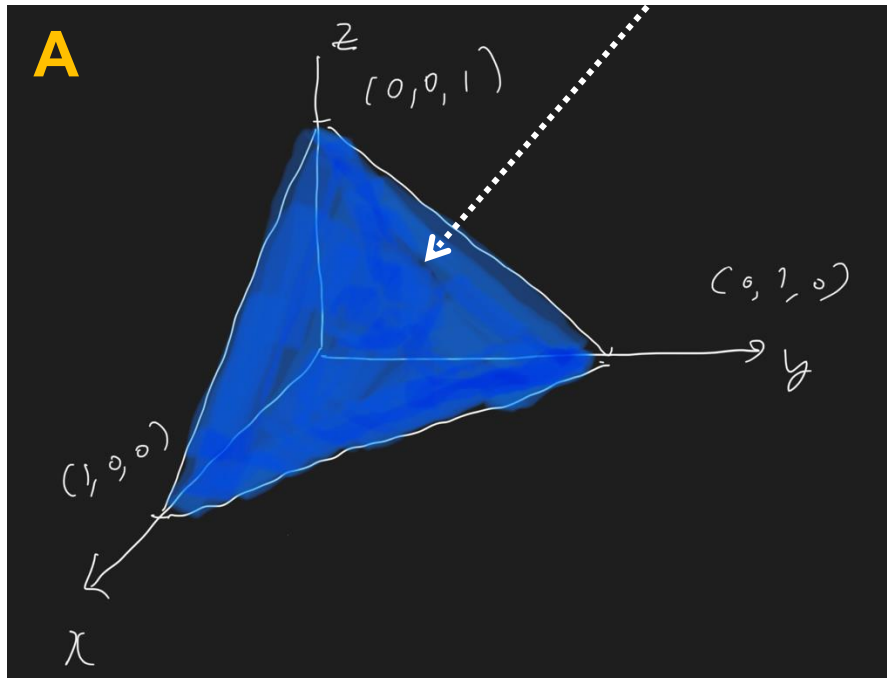


Chung et al. 2003 NeuroImage

Chung et al. 2008. IEEE Transactions on Medical Imaging

Exercise: probability distribution on 2-simplex

$$f(x, y, z)$$



Dirichlet distribution on triangle A

$$f(x_1, x_2, x_3) = \frac{\Gamma(\alpha_1 + \alpha_2 + \alpha_3)}{\Gamma(\alpha_1)\Gamma(\alpha_2)\Gamma(\alpha_3)} x_1^{\alpha_1-1} x_2^{\alpha_2-1} x_3^{\alpha_3-1}$$

Problem. Find a distribution on triangle B.

Geodesics: the shortest path on a surface

A geodesic is a curve that locally minimizes length — the “straightest” possible path on a curved surface.

On a curved surface, “straight lines” are called **geodesics**!



1 Length of a curve on a surface

Let $X(u^1, u^2)$ be a parametrization of a surface with metric g_{ij} . For a curve $\gamma(t) = X(u^1(t), u^2(t))$, its length is

$$L[\gamma] = \int_a^b \sqrt{g_{ij} \dot{u}^i \dot{u}^j} dt$$

Geodesics are the shortest paths on the surface!



A geodesic is a curve that locally minimizes this length (a locally shortest path).

2 Geodesic equation

The geodesics satisfy the Euler–Lagrange equations for the length functional, which give

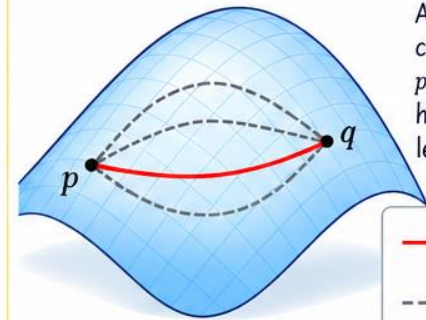
$$\frac{d^2 u^k}{dt^2} + \Gamma_{ij}^k \frac{du^i}{dt} \frac{du^j}{dt} = 0 \quad (k = 1, 2)$$

where the coefficients Γ_{ij}^k are determined by the metric:

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$$

Solving this second-order ODE (with given endpoints) gives the geodesic curve $u^1(t), u^2(t)$.

3 Geodesic is locally shortest



Among all nearby curves connecting p to q , the geodesic has the shortest length (locally).

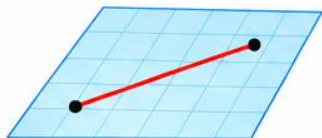
— geodesic (shortest)
--- other path (longer)

4 Examples

a Plane (Euclidean)

Metric: $g_{ij} = \delta_{ij}$

Geodesics: straight lines

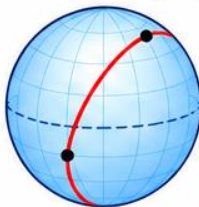


On a flat plane, geodesics are the usual straight lines.

b Sphere (radius R)

Metric: $g_{ij} = R^2 \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}$

Geodesics: great circles

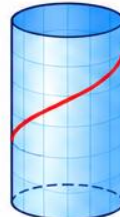


On a sphere, the shortest path between two points is an arc of a great circle (e.g. part of the equator).

c Cylinder (radius R)

Metric: $g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & R^2 \end{pmatrix}$

Geodesics: helices or straight lines



On a cylinder, geodesics are helices (slanted straight lines when unrolled), and also horizontal/vertical lines.

5 Finding the shortest path (procedure)

- 1 Compute the metric g_{ij} for the surface.
- 2 Solve the geodesic equation $\frac{d^2 u^k}{dt^2} + \Gamma_{ij}^k \frac{du^i}{dt} \frac{du^j}{dt} = 0$ with boundary conditions (endpoints).
- 3 Among the solutions, choose the one with minimal length (usually the geodesic is already the minimizer locally).

6 Beyond 2D: on any manifold

The same idea works on higher-dimensional manifolds: geodesics are curves that locally minimize the length defined by the metric g_{ij} .

Geodesics show us the natural straightest paths in curved spaces!

Geodesics generalize straight lines to curved surfaces!



Mean and Gaussian curvatures

Curvature measures how the surface bends around each point!



1 Surface and metric

Let $X(u^1, u^2)$ be a local parametrization of a surface $M \subset \mathbb{R}^3$. The metric tensor (first fundamental form) is

$$g_{ij} = \langle \partial_i X, \partial_j X \rangle, \quad i, j = 1, 2$$

where $\partial_i = \frac{\partial}{\partial u^i}$ and $\langle \cdot, \cdot \rangle$ is the Euclidean inner product in $\mathbb{R}^3$.

The metric encodes all intrinsic geometry: lengths, angles, and area on the surface.

2 Unit normal and second fundamental form

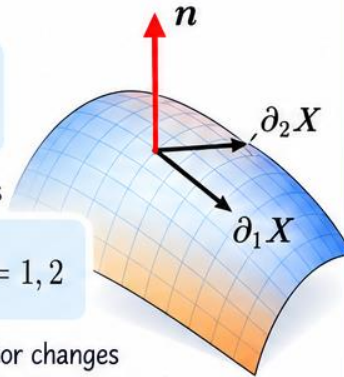
The unit normal vector is

$$n = \frac{\partial_1 X \times \partial_2 X}{\|\partial_1 X \times \partial_2 X\|}$$

The second fundamental form is

$$b_{ij} = \langle \partial_i \partial_j X, n \rangle, \quad i, j = 1, 2$$

It measures how the normal vector changes as we move on the surface (extrinsic geometry).



3 Mean and Gaussian curvatures

Let g^{ij} be the inverse matrix of g_{ij} . Then

$$H = \frac{1}{2} g^{ij} b_{ij} = \frac{1}{2} (\kappa_1 + \kappa_2) \quad \text{Mean curvature}$$

$$K = \frac{\det(b_{ij})}{\det(g_{ij})} = \kappa_1 \kappa_2 \quad \text{Gaussian curvature}$$

where κ_1, κ_2 are the principal curvatures (eigenvalues of the shape operator).

4 Principal curvatures

In an orthonormal basis of principal directions e_1, e_2 , the forms are diagonal:

$$g_{ij} = \delta_{ij}, \quad b_{ij} = \begin{pmatrix} \kappa_1 & 0 \\ 0 & \kappa_2 \end{pmatrix}$$

so that $H = \frac{1}{2} (\kappa_1 + \kappa_2)$, $K = \kappa_1 \kappa_2$.



H tells the average bending, K tells how the surface bends in total!

5 Examples

Sphere (radius R)

$$X(\theta, \phi) = \begin{pmatrix} R \sin \theta \cos \phi \\ R \sin \theta \sin \phi \\ R \cos \theta \end{pmatrix}$$



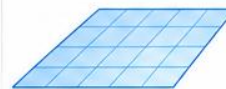
$$g_{ij} = \begin{pmatrix} R^2 & 0 \\ 0 & R^2 \sin^2 \theta \end{pmatrix}$$

$$H = \frac{1}{R}, \quad K = \frac{1}{R^2}$$

Both positive (curves like a sphere).

Plane

$$X(u, v) = (u, v, 0)$$



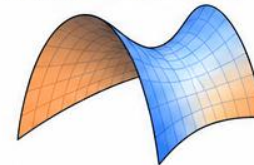
$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$H = 0, \quad K = 0$$

Flat (no curvature).

Saddle (hyperbolic paraboloid)

$$X(u, v) = (u, v, u^2 - v^2)$$



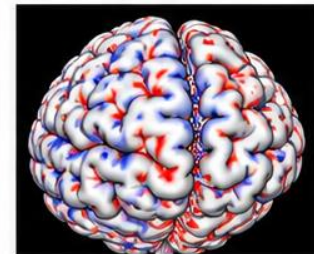
$$g_{ij} = \begin{pmatrix} 1 + 4u^2 & -4uv \\ -4uv & 1 + 4v^2 \end{pmatrix}$$

$$H = 0, \quad K < 0$$

Saddle shape (opposite curvatures).

6 Brain surface example

The cerebral cortex is a highly folded surface. Regions with positive K are locally spherical (gyrus), negative K are saddle-like (sulcus).



$K > 0$ (sphere-like)
0
 $K < 0$ (saddle-like)

Mean and Gaussian curvatures help quantify and analyze the shape of the brain surface.



Unit Outcomes

- 1) Understand curve and surface parametrization
- 2) Understand Riemannian metric tensors
- 3) Able to compute curvatures